# Learning as optimisation

## AI · University · Lecture 1



# In this lecture

- Frame learning as minimising empirical risk.
- Run gradient descent and SGD.
- Diagnose learning-rate problems.

# Remember

- $\theta \leftarrow \theta - \eta \nabla \hat{R}$.
- Too big a step diverges.
- The gradient is local, not a map.

# Steps by hand

1. Warm-up (5 min): Where is the minimum?
2. Derive (15 min): The update rule.
3. Step (15 min): Three rates.
4. Code (20 min): SGD.
5. Wrap-up (5 min): One-minute paper.

# Let's discuss

1. At which $\eta$ did $\theta^2$ diverge, and why?
2. Why does SGD jitter?
3. How will you assess curve reading?

# Exit ticket

- Sketch the loss curve for a learning rate that is too big, and label it.